



Iowa City Robotics

Catch a Rainbow

Summary

Students explore how soap affects liquids by experimenting with milk, food coloring, and dish soap.

Description

Students will observe how drops of food coloring move differently in water and milk. When dish soap is added, the colors quickly swirl and move across the milk. This simple experiment helps students see how soap changes how liquids behave.

Materials

Material	Quantity
Shallow bowl	1 per group of 2
Food coloring (multiple colors)	3–4 bottles per classroom
Deep bowl for water experiment	1
Liquid dish soap	1 bottle
Whole milk (or buttermilk)	About ½ gallon per 100 groups
Toothpicks	1 box

Optional for extension: skim milk, cream, or other dairy products

Directions

- Fill the deep bowl with water, almost to the top
- Add one drop of food coloring to the water and watch what happens
- Pour enough milk into a shallow bowl to cover the bottom (about 1 cm deep)
- Add 2–3 drops of different food coloring to the milk
- Observe how the colors move in the milk
- Dip a toothpick into dish soap and gently touch the center of the milk
- Watch what happens to the colors
- Compare what happened in the water and what happened in the milk



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Activity Extension

- Try the same experiment with different dairy products such as skim milk or cream
- Ask students to predict what will happen before adding the soap
- Let students experiment with adding soap in different places in the bowl

Discussion Questions

- What happened to the food coloring in the water?
- What happened to the food coloring in the milk?
- Did the colors move slowly or quickly?
- What happened when soap touched the milk?
- Where did the colors move?
- Why do you think the colors moved away from the soap?

What is Happening?

- Food coloring spreads easily in water because the water and dye mix well together
- In milk, the dye moves more slowly because milk contains fats
- When dish soap is added, it breaks the surface tension of the milk
- The soap also interacts with the fats in the milk, which causes the colors to swirl and move
- The movement you see is caused by the liquid mixing and flowing, which creates the colorful rainbow patterns